Jiuchen Zhang

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Research Interests

Methodology: Statistical and machine learning methods for high-dimensional and longitudinal data; neural networks and data-adaptive models with valid inference

Applications: Exposome modeling and environmental mixtures; life-course and child growth trajectories; environmental health and mobile/wearable health studies

Academic & Professional Experience

Jul. 2024 – Present
Postdoctoral Research Fellow, Department of Biostatistics
University of Michigan, Ann Arbor, MI
Advisor: Peter X.-K. Song
Develop statistical learning methods for neural-network inference, longitudinal and high-dimensional data, and environmental-health applications.

Dec. 2021 – Jun. 2024
Research Assistant, Dementia Family Caregiver Support Study
University of California, Irvine, CA
Managed and analyzed wearable-device data; developed automated data pipelines; and supported interdisciplinary manuscripts and conference abstracts.

May 2019 – Dec. 2019
Statistical Consultant, Illinois Statistics Office
University of Illinois at Urbana-Champaign, Champaign, IL
Advised interdisciplinary teams on study design, power analysis, data management, and statistical methodology.

Sep. 2018 – May 2019
Statistical Analyst, Technical Consulting & Research, Inc., Weston, CT
Developed Python data-acquisition pipelines and performed data cleaning, record linkage, and geocoding for clinical-trial data.

Education

2020 – 2024 **University of California Irvine** – Irvine, CA
PhD in Statistics
Advisor: Dr. Annie Qu

2016 – 2018 **University of Illinois at Urbana-Champaign** – Champaign, IL
Master of Science in Statistics

2012 – 2016 **Zhejiang University** – Hangzhou, Zhejiang
Bachelor of Science in Mathematics and Applied Mathematics

Publications & Manuscripts

Published & Accepted

1. **Zhang, J.**, Nie, R., & Song, P. X.-K. (2026). Comments on "An Overview of Large Language Models for Statisticians." *The American Statistician.* Accepted discussion.

2. Kim, E. A., **Zhang, J.**, Rahmani, A. M., Shin, S., Nyamathi, A., & Lee, J.-A. (2026). Unsupervised learning identifies sleep disturbance subtypes among dementia caregivers. *Nursing Research.* Accepted.

3. **Zhang, J.**, Xue, F., Xu, Q., Lee, J.-A., & Qu, A. (2024). Individualized dynamic latent factor model for multi-resolutional data with application to mobile health. *Biometrika, 111*(4), 1257–1275.

4. **Zhang, J.**, Yuan, Y., & Qu, A. (2022). Tensor factorization recommender system with dependency. *Electronic Journal of Statistics, 16*(1), 2175–2205.

Published Conference Abstracts

5. Lee, J.-A., **Zhang, J.**, Rahmani, A., & Qu, A. (2024). Wearable Internet-of-Things technology to measure sleep of diverse caregivers: Preliminary results from a randomized controlled trial. *Alzheimer's & Dementia, 20*, e092479. Published conference abstract.

6. Lee, J.-A., **Zhang, J.**, Sabino-Laughlin, E., Ju, E., Rahmani, A., & Qu, A. (2023). Improvements in heart rate variability among dementia family caregivers receiving a home-visit based intervention: Preliminary results from a randomized controlled trial. *Alzheimer's & Dementia, 19*, e077110. Published conference abstract.

7. Lee, J.-A., Ju, E., **Zhang, J.**, Sabino-Laughlin, E., Rahmani, A., Gibbs, L., Shin, S., Qu, A., Kehoe, P., & Nyamathi, A. (2022). Culturally and linguistically appropriate home visit intervention for underserved dementia family caregivers: Preliminary results from a randomized controlled trial. *Alzheimer's & Dementia, 18*, e058927. Published conference abstract.

Public Preprints

8. **Zhang, J.**, Zhou, L., & Song, P. X.-K. (2026). Neural network machine regression (NNMR): A deep learning framework for uncovering high-order synergistic effects. *arXiv preprint*, arXiv:2602.02172.

Manuscripts Under Review and Revision

9. **Zhang, J.**, & Qu, A. (Revision submitted). Testing cumulative treatment effects in continuous time with ODE dynamics and noisy observations. *Journal of the American Statistical Association*. Revision submitted following a reject-and-resubmit decision; arXiv:2602.02246 (earlier version under a different title).

10. **Zhang, J.**, & Song, P. X.-K. (Under review). A smooth low-rank time basis for neural estimation of cumulative longitudinal means. *International Conference on Learning Representations (ICLR 2027)*.

11. **Zhang, J.**, Nie, R., & Song, P. X.-K. (Under review). VEA-aware uncertainty for LLM chain of actions. *International Conference on Learning Representations (ICLR 2027)*.

12. Chen, C., Nie, R., Yi, M., Zeng, Q., **Zhang, J.**, Zhang, Y., & Song, P. X.-K. (Major revision). New advances on computational toolboxes and software for six essential statistical learning methods: A systematic review. *WIREs Computational Statistics*.

13. Chen, J., **Zhang, J.**[*], Tang, N., & Qu, A. (Under review). Direct multiple function-on-function regression under multi-resolution sampling: Risk decomposition and structural inference. *Journal of the American Statistical Association*.
[*]Corresponding author.

14. Ye, H., **Zhang, J.**, & Qu, A. (Under review). Multi-task active learning with efficient resource allocation. *Journal of the American Statistical Association*. arXiv:2610.07045.

15. Pan, J., **Zhang, J.**, Zheng, Z., Song, P. X.-K., Colacino, J. A., Peterson, K. E., & Dolinoy, D. C. (Under review). Effects of perinatal lead exposure with antioxidant diet supplementation on murine offspring growth and metabolic outcomes in adulthood. *Toxicological Sciences*.

Manuscripts in Preparation

16. **Zhang, J.**, Zhang, K., & Song, P. X.-K. Likelihood-based inference for binary decision trees via the binary expansion framework. Under internal revision; planned submission by the end of 2026.

17. Yi, M.[†], **Zhang, J.**[†], Wang, H. J., & Song, P. X.-K. Projection-powered quantile regression with auxiliary data and transfer learning. Planned submission by the end of July 2027.
[†]Co-first authors.

Grant Applications

Feb. 2026 **NIH K99/R00 Pathway to Independence Award Application**
Inference-Oriented Neural Network Modeling of Early-Life Exposome Effects on Child Growth Trajectories
PD/PI; submitted February 2026; reviewed June 2026; impact score: 49.

Software

2024 **IDLFM: Individual Dynamic Latent Factor Model**
Siyang Liu, Jiuchen Zhang, Annie Qu.
R package available on CRAN; nearly 3,000 downloads since May 2025.
DOI: 10.32614/cran.package.idlfm.

Honors & Awards

2024 First Place, ASA MDD Student Paper Competition (JSM 2024)

2024 ENAR Distinguished Student Paper Award (ENAR Spring Meeting)

2013 Second-Class Scholarship for Elite Student in Basic Sciences (Zhejiang University)

2013 Third-Class Scholarship for Outstanding Merit (Zhejiang University)

Teaching Experience

Invited Guest Lecturer, Sue & Bill Gross School of Nursing, University of California, Irvine

2025 Spring — NS 213: Measurements (PhD course) — Introduction to Longitudinal Data Analysis

Teaching Assistant, University of California, Irvine

2021 Fall — STATS 200AP: Intermediate Probability and Statistical Theory I (developed course slides)

2021 Spring — STATS 7: Basic Statistics (two sections)

Mentoring

2023 — Haoqun Cao — UCInspire: Semi-Markov Modeling and Change-Point Detection for Maternal Sleep-Stage Trajectories

Xieqing Yu — UCInspire: Semi-Markov Proportional-Hazards Modeling of Maternal Sleep-Stage Trajectories

2022 — Zhuojiao Dong — STATS 199: Deep Learning Models for Health and Stress Prediction

Yiting Wang — Dementia Family Caregiver Support Study: Data Management

Yiheng Wei — Dementia Family Caregiver Support Study: Garmin Wearable-Data Analysis

Haochen Tan — STATS 199: Deep Learning Models for Health and Stress Prediction

Ruxi Zheng — STATS 199: Deep Learning Models for Health and Stress Prediction

Presentations

Aug. 2025 — A Scaling Neural Network-Based Approach for Variable Selection in Nonparametric Regression
2025 Joint Statistical Meetings (JSM)

Aug. 2024 — Individualized Dynamic Latent Factor Model for Multi-resolutional Data with Application to Mobile Health
2024 Joint Statistical Meetings (JSM)

Mar. 2024 — Individualized Dynamic Latent Factor Model for Multi-resolutional Data with Application to Mobile Health
2024 Eastern North American Region (ENAR) Spring Meeting

Aug. 2023 — Individualized Dynamic Model for Multiresolutional Data
2023 Joint Statistical Meetings (JSM)

Sep. 2021 A Tensor Factorization Recommender System with Dependency
Invited talk, 2021 ICSA Applied Statistics Symposium

Aug. 2020 Multilayer Recommender Systems Using Dependent Tensor
2020 Joint Statistical Meetings (JSM)

Professional Service

2025 **Session Chair**, "Harnessing the Power of Large-Scale and Heterogeneous Data with Integrative Analysis"
Joint Statistical Meetings (JSM)

Peer Review

Electronic Journal of Statistics (3 reviews)

Journal of the American Statistical Association (3 reviews)

Journal of Multivariate Analysis (1 review)

Journal of the Royal Statistical Society: Series B (3 reviews)

ACM Transactions on Knowledge Discovery from Data (1 review)

Scientific Reports (1 review)

Professional Membership

American Statistical Association

International Chinese Statistical Association

Institute of Mathematical Statistics

Eastern North American Region International Biometric Society